

Linear Data Pipeline

Executive Overview

How we copy, store, and use Linear data in Google Cloud

Prepared for:	Leadership review
GCP project:	peq-tools (non-production)
Status:	Phase 3 complete - building toward full pipeline
Date:	June 2026

Visual guide - minimal technical jargon

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1. Executive summary

We are building a reliable copy of our Linear (project management) data inside Google Cloud. Once complete, reports and tools will read from our own database instead of calling Linear directly every time.

Bottom line

Phase 3 is complete: we can copy Linear data into secure cloud storage on demand. The daily sweep email already works on weekdays at 9 AM. Next steps turn that archive into queryable tables and point the email at those tables.

In one sentence

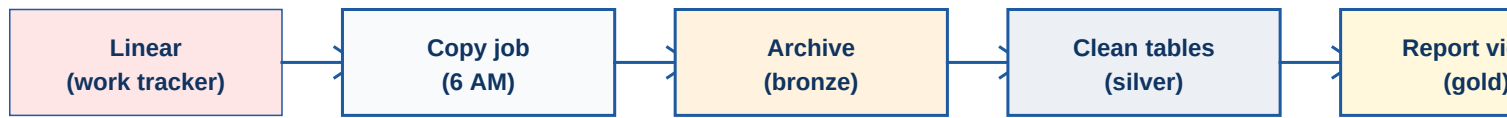
Linear remains the source of truth; Google Cloud becomes our durable, auditable copy that powers internal reporting.

2. Why we are building this

- Auditability - prove what we knew on any past date (immutable archive)
- Reliability - reports do not break if Linear is slow or rate-limits us
- Speed - query our database instead of re-downloading history daily
- Foundation - same pattern works for Epic and other sources later
- Security - credentials in Google Secret Manager, not in scripts or inboxes

3. How the pipeline works (visual)

End-to-end flow (target state)

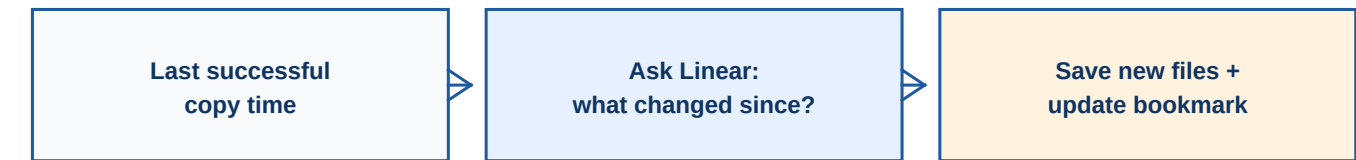


Think of it like a photocopier (copy job) that files snapshots (archive), then a librarian sorts them into a catalog (silver), and reports read from a simple summary shelf (gold).

The three storage layers

BRONZE - Raw archive Exact copies as JSON files in cloud storage. Never deleted - full history for audits.	Done today
SILVER - Master tables One current row per issue, user, team, etc. Updates merge in - no duplicates.	Phase 4
GOLD - Report-ready views Simple views shaped for emails and dashboards. Tools only read from here.	Phase 5

How "only new changes" works (bookmark)



After the first full copy (backfill), daily runs only fetch records that were updated since the last bookmark - not the entire history every morning.

4. What is done vs. what is next

Project progress

Phase 1-2	Platform + daily email		Done
Phase 3	Copy Linear to cloud archive		Done
Phase 4	Clean data in BigQuery (silver)		Planned
Phase 5	Reports use database, not Linear API		Planned
Phase 6	Monitoring, retention, production org		Planned

Completed (validated)

- Cloud infrastructure in peq-tools (storage, database shells, jobs)
- linear-ingest copy job - local and cloud tested
- Full backfill of test Linear workspace into cloud archive
- Bookmark file tracks last copy time per data type
- Weekday 9 AM sweep email sending successfully

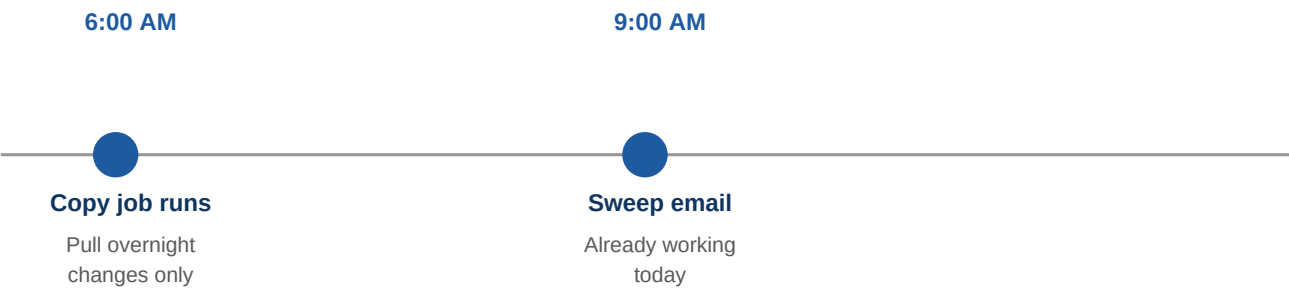
Not started yet

- Silver tables - cleaned master data in BigQuery
- Gold views - report-shaped data for tools
- Sweep email reading from database instead of Linear API
- Automatic 6 AM daily copy schedule
- Production Linear organization backfill

5. Daily schedule

Today only the 9 AM email is on a timer. The 6 AM copy job will be scheduled after silver/gold layers are built - so data is fresh before the email goes out.

Typical weekday schedule (target)



6. Security & the API key

To copy production Linear data we need a read API key from the real organization. It will be stored once in Google Secret Manager and used only by the automated copy job.

Where the API key lives



- Key is never stored in code, email, or spreadsheets
- Separate secret for the copy job vs. the email job (planned split)
- Access is logged; only approved cloud accounts can read secrets

What the key is used for

- Read-only copy of issues, users, teams, comments, and related records
- Not shared with the email job once we finish Phase 5 (email uses database only)

What we are NOT doing

- Storing the key in GitHub, code, or Terraform
- Giving every employee direct access to the key
- Writing changes back into Linear

7. Audit & record-keeping

Every copy run creates a new dated folder in cloud storage. We do not overwrite historical snapshots - we add new ones. That supports questions like "What did Linear show us on March 3?"

What is logged today

- Each copy run: timestamp, record counts, success/failure per data type
- Cloud job execution logs in Google Cloud Logging
- Secret access audit trail for API key reads

Planned improvements

- Retention policy (how long to keep raw archives)
- Alerts if copy or email jobs fail
- Master tables in BigQuery with merge history

8. What we need to go live on real data

To move from the small test workspace to production Linear:

- Production Linear API key (read access) added to ingest secret
- One-time full copy (backfill) of the real organization - may take longer than test
- Complete Phase 4-5 before reports depend on database copy
- Confirm which teams/issues are in scope for reporting

Recommendation

Provide the production API key when ready for backfill. We can run a dry-run first (connectivity check, no storage changes) before the full copy.

9. FAQ

Q: Is the daily email already working?

A: Yes - weekday 9 AM emails are sending. They still call Linear directly today; the database pipeline will replace that later.

Q: Is data copying automatically every morning?

A: Not yet - copy runs on demand. A 6 AM schedule comes after silver/gold are built.

Q: Will we have a big -bronze database- in BigQuery?

A: The raw archive lives in cloud file storage (bronze). BigQuery will hold cleaned silver tables and gold report views - not necessarily another bronze copy.

Q: Does incremental copy re-download everything?

A: No - after the first full copy, daily runs use a bookmark to fetch only changes.

Q: What if we add Epic or other tools later?

A: Same pattern: separate copy job, archive, and silver/gold per source. Cross-tool reports join in a shared analytics layer.

Q: How much will this cost?

A: At our scale, storage and queries in Google Cloud are expected to be low (typically dollars per month, not thousands) for Linear-sized data.